AI and Robotics

Lesson 2b: Layered Decision-Making in Robotics

Sarah Keren

The Taub Faculty of Computer Science
Technion - Israel Institute of Technology

TECHNION | The Henry and Marilyn Taub Faculty of Computer Science

Source

Probabilistic Robotics
S. Thrun, W. Burgard, and D. Fox, Probabilistic Robotics.
Cambridge, MA: MIT Press, 2005.
Chapters 2-3

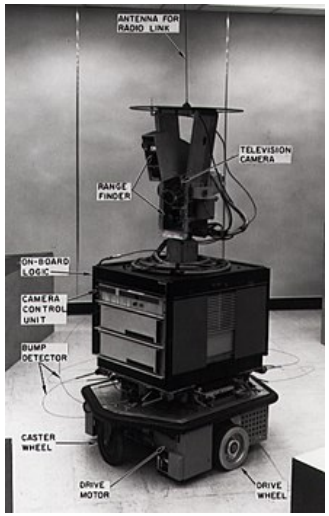
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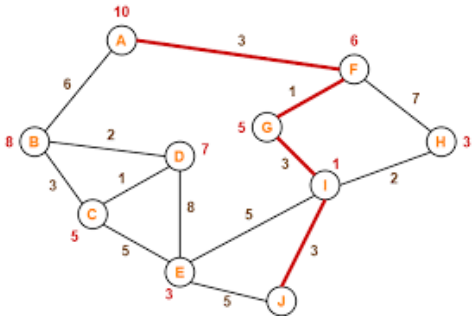
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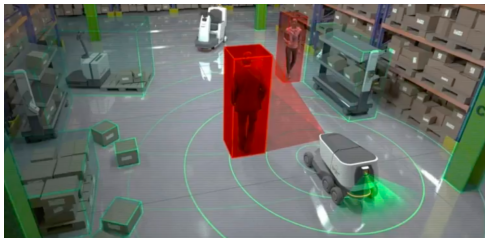
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Layered Decision Making

What goes into 'go from A to B'?





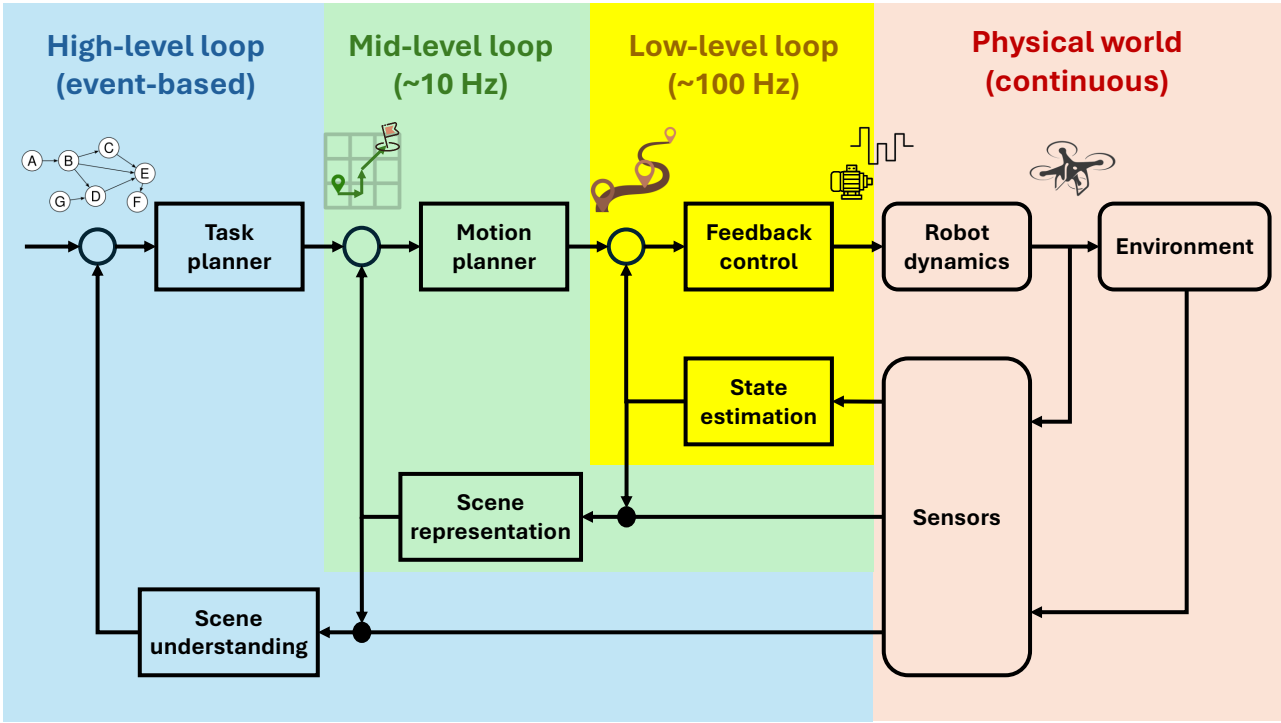


What goes into 'pick up an apple'?



What about: "I am hungry" ?

Layered Decision-Making

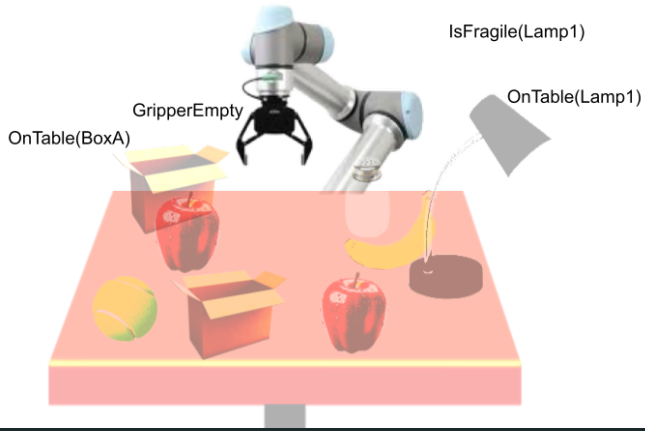


We will consider two main elements: state estimation and decision-making (and their relationship).

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Task Planning

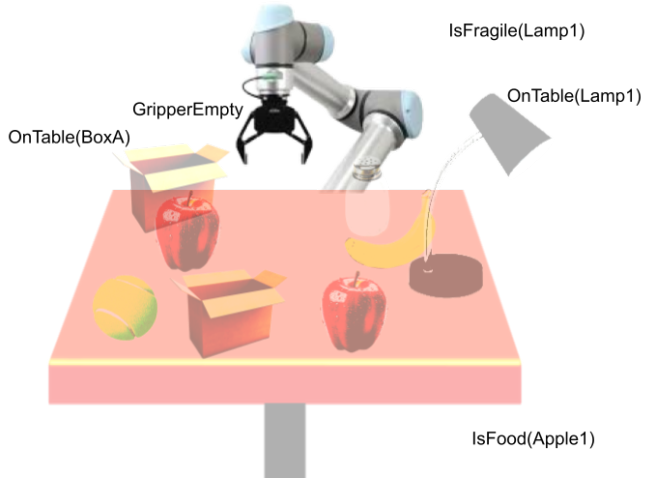
Let $\mathcal{R}$ be a robot operating in a discrete environment described by a set of symbolic **states** $\mathcal{S}$ and **actions** $\mathcal{A}$. Given an initial state $s_0 \in \mathcal{S}$ and a **goal condition** $G \subseteq \mathcal{S}$, determine whether there exists a sequence of actions $\pi = \langle a_1, a_2, \ldots, a_n \rangle$, such that executing π from s_0 leads to a state $s_n \in G$.



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Task Planning

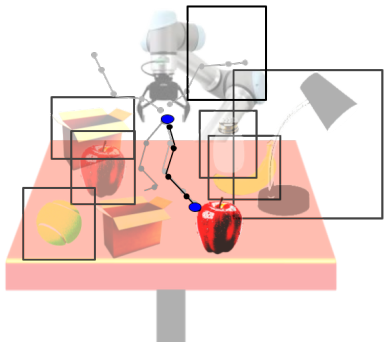
- **Input:** Goal condition: 'pick up an apple'
- **Output:** A sequence of (high-level) actions that achieve the goal
- An option: use propositional STRIPS to define the planning task as $P = \langle \mathcal{F}, I, \mathcal{A}, G, \mathcal{C} \rangle$ and use A* to solve the problem



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Motion Planning

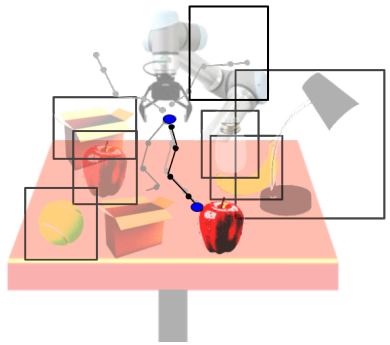
Let $\mathcal{R}$ be a robot system with d **degrees of freedom**, moving in a known environment with obstacles. Given start and target **configurations** q_0 and q_g for $\mathcal{R}$, decide whether there is a collision-free, continuous path $\tau : [0, 1] \to \mathcal{C}_{free}$ such that $\tau(0) = q_0$ and $\tau(1) = q_g$ and if so, plan such a motion.



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Motion Planning

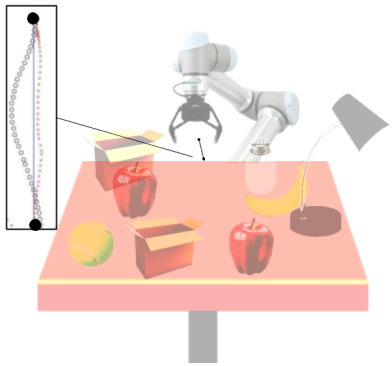
- **Input:** Initial and goal configurations and map of (obstacles in the) environment.
- **Output:** A collision-free path from the initial to the goal configuration.
- An option: map the environment into free and occupied space and use A* over the free space.



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Motion Control

Given a desired trajectory $\tau(t)$ for a robotic system $\mathcal{R}$ with d **degrees of freedom**, the goal of motion control is to compute and apply control inputs $\mathbf{u}(t)$ (e.g., joint torques, velocities, or forces) such that the system's actual motion $q(t)$ follows $\tau(t)$ as closely as possible.



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Motion Control

- **Input:** Desired trajectory
- **Output:** A control procedure that minimizes the deviation between the executed and desired trajectory.

System dynamics are captured by $\dot{x}(t) = f(x(t), u(t), t) + w(t)$ where

- $x(t) \in \mathbb{R}^n$ - state vector
- $u(t) \in \mathbb{R}^m$ - control input
- $f : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \to \mathbb{R}^n$ - system dynamics
- $w(t)$ - noise or disturbance

The challenge is to translate actions into low-level commands (and monitor their execution).

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From Instructions to Torques

'pick up an apple'

- `pick_up_apple`
- `move_to_grasp_config`
- `follow_trajectory`

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From Instructions to Torques

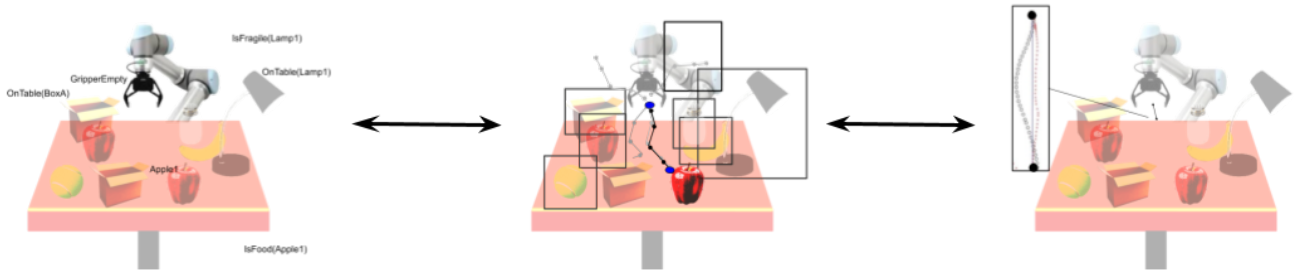
'pick up an apple'

- `pick_up_apple`
  - **State Estimation:** locate an apple in the scene
  - **Decision Making:** decide which apple to try and pick up
- `move_to_grasp_config`
  - **State Estimation:**
    offline: mark all the obstacles in the environment (including the robot) in the configuration space.
    online: detect robot position (in dynamic environments, also the position of obstacles)
  - **Decision Making:** decide how to reach the configuration
- `follow_trajectory`
  - **State Estimation:** monitor robot location w.r.t. reference trajectory
  - **Decision Making:** corrective actions

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Open Questions

- How to update the state representation?
- How to interface the three layers?
  - For example, which component maintains valid grasp configurations?
- How to account for uncertainty?



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Modeling Uncertainty

Uncertainty in Robotics

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Uncertainty in Robotics

- Robot environments are inherently unpredictable (especially if people are around)
- Sensors are limited in what they can perceive
- Actuators [somewhat] unpredictable (especially for low-cost and mobile robots)
- Modeling uncertainty - all models are abstractions
- Algorithmic approximation

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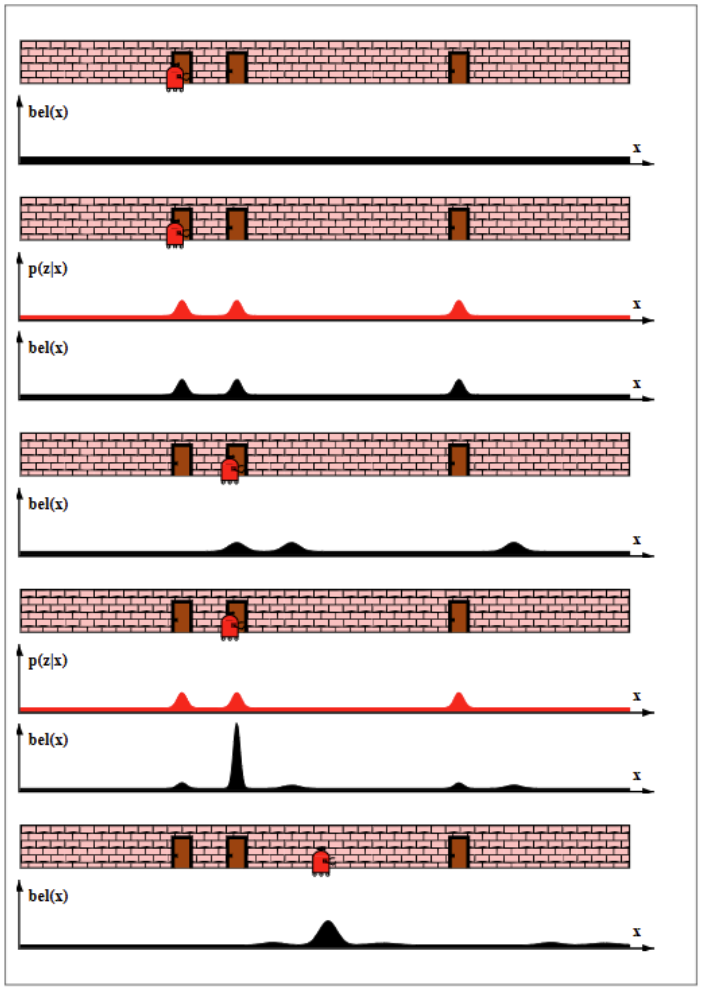
Uncertainty in Robotics

- We are looking for ways to explicitly account for uncertainty, instead of relying on a single ”best guess”.
- We want decisions to be robust
- We want robots to actively choose to reduce uncertainty when this is the superior choice.

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Example - Markov Localization

The agent is given a map of an environment with 3 indistinguishable doors.
Initially, it does not know where it is and needs to localize itself on the map using its sensor data.



From Probabilistic Robotics S. Thrun, W. Burgard, and D. Fox, Probabilistic Robotics. Cambridge, MA: MIT Press, 2005.

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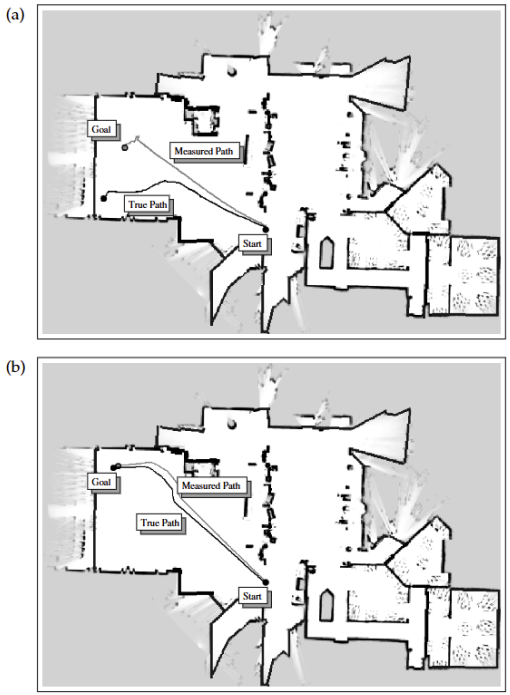
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Coastal Navigation

In addition to accounting for monetary uncertainty, an agent can anticipate future uncertainty and plan accordingly.
Estimated and actual paths differ when the agent uses features (or landmarks) from the environment.



By Nicholas Roy MIT

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What are the models we know for modeling uncertainty?

Reminder: Markov Decision Process (MDP)

A **Markov Decision Process** (MDP) is a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ where

- $\mathcal{S}$ is a finite set of states (states are described via a set of random variables $\mathcal{X} = X_1, \ldots, X_n$).
- $\mathcal{A}$ is a finite set of actions
- $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ is the **transition probability function**, where $\mathcal{P}(s' \mid s, a) = p(S_{t+1} = s' \mid S_t = s, A_t = a)$ denotes the probability of transitioning to state s' after taking action a in state s.
- $\mathcal{R} : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the **reward function**, where $\mathcal{R}(s, a)$ (or its expectation) gives the immediate reward obtained after taking action a in state s, and
- **optional:** γ is a discount factor $\gamma \in [0, 1]$ that is used to favor immediate rewards over future rewards.

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Reminder: Partially Observable Markov Decision Process (POMDP)

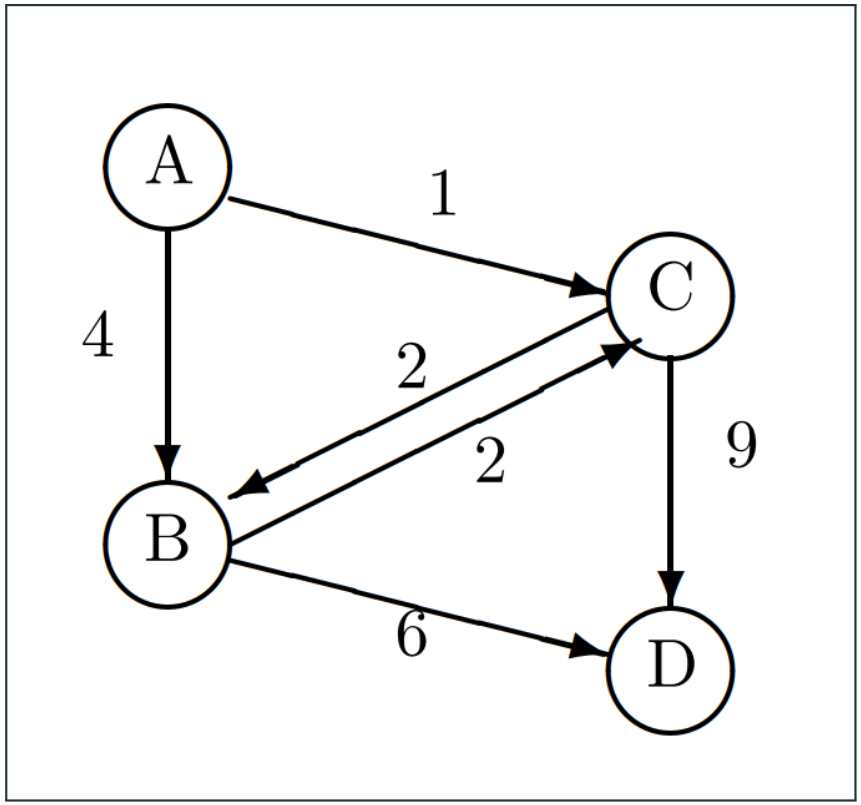
A **Partially Observable Markov Decision Process**(POMDP) is a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, \Omega, \mathcal{O}, \beta_0 \rangle$ where

- $\mathcal{S}$, $\mathcal{A}$, $\mathcal{P}$, $\mathcal{R}$ and γ are as for an MDP.
- Ω is a set of observations (observation tokens),
- $\mathcal{O} : \Omega \times \mathcal{S} \times \mathcal{A} \to [0, 1]$ is a sensor function specifying the conditional probability $\mathcal{O}(o|s, a) = p(O_{t+1} = o|S_t = s, A_t = a)$ of receiving observation $o \in \Omega$ in state s after applying a [1].
- β_0 the initial belief: a probability distribution over the states.

[1] alternatively: $\mathcal{O}(o|s) = p(O_t = o|S_t = s)$

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We know that the model for A* and best-first-search is a graph.

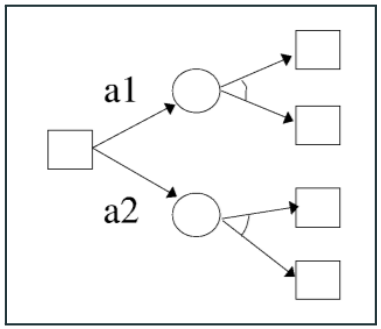


What is the underlying model for an MDP and POMDP?

And-Or Graphs

- For conditional / probabilistic planning we need to take some action at every state, but must handle every outcome for the action.
- A solution is a conditional plan / policy rather than just a single move.
- For this purpose we use and-or-graphs with two kinds of nodes:
  - **OR node** - specify a selection between actions
  - **AND node** - specify the possible outcomes

Where have you already used an and-or graph ?



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And-Or Graphs

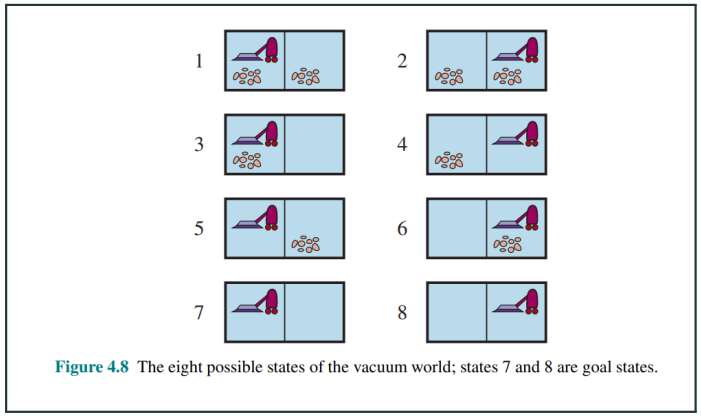


Figure 4.8 The eight possible states of the vacuum world; states 7 and 8 are goal states.

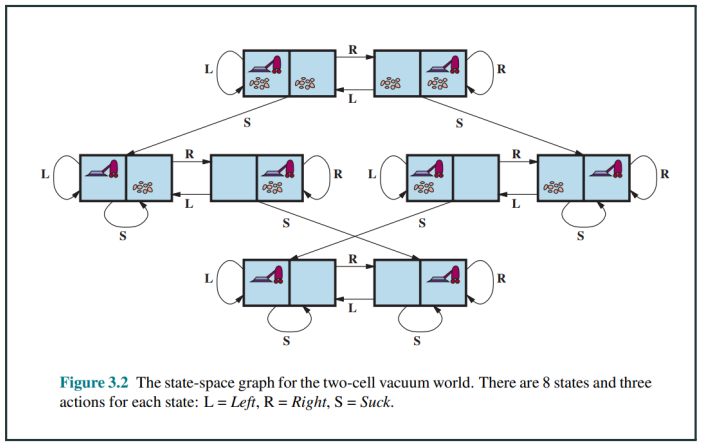


Figure 3.2 The state-space graph for the two-cell vacuum world. There are 8 states and three actions for each state: L = *Left*, R = *Right*, S = *Suck*.

From Artificial Intelligence A Modern Approach by Russel and Norvig

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And-Or Graphs

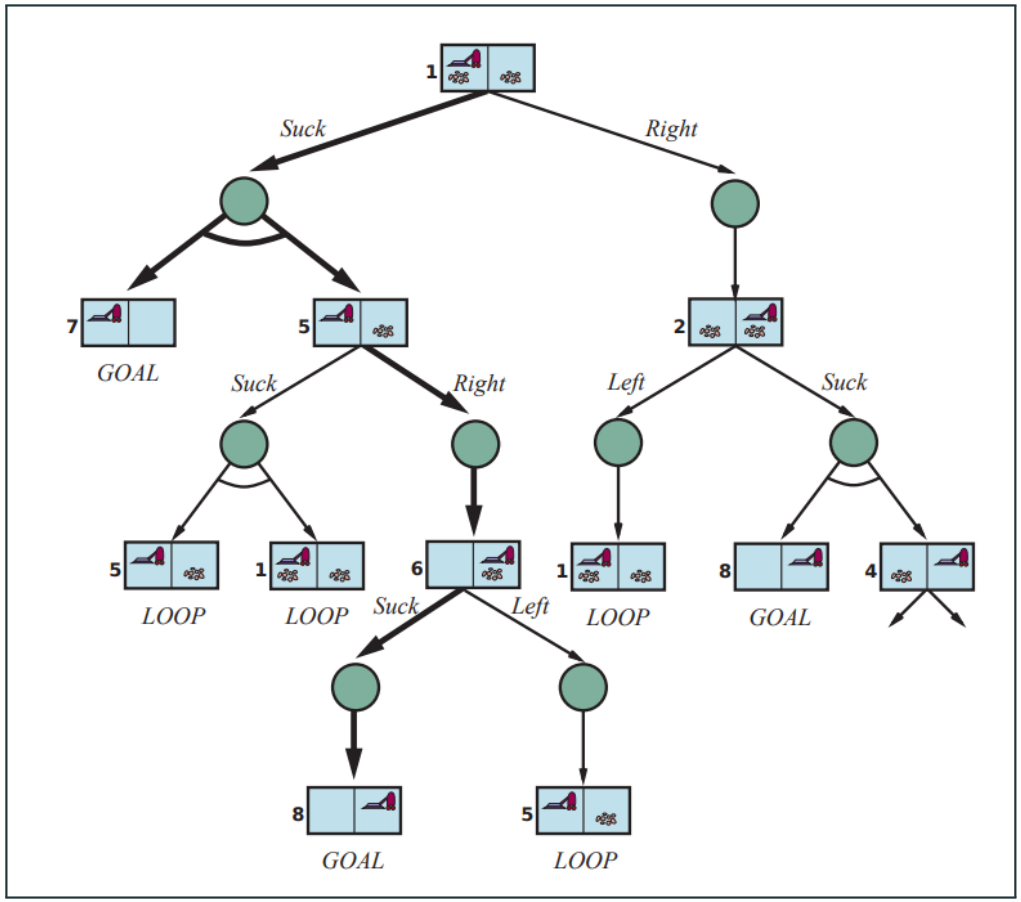
The setting has various versions:

- Fully observable, deterministic, static, discrete. The agent perceives only the current location and whether it's dirty. Dirt does not reappear.
- Partially observable, deterministic, static, discrete. The agent cannot sense the [other] square.
- Partially observable, stochastic, dynamic. Dirt can reappear randomly with some probability.

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And-Or Graphs

Which setting does this correspond to ?



How do we account for partial observability (instead of no-sensing)?

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Probabilistic Transition and Sensor Functions

- Most commonly, the transition and sensor functions are modeled as a probability distribution.
- Sometimes, the reward function is modeled as a distribution.

What's missing?

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Probability 101

Basics Concepts in Probability

- Random variable - takes values according to specific probabilistic laws
  - In robotics: sensor measurements, controls, states
- Probabilistic inference - the process of calculating these laws for random variables based on other random variables and observations.
- We denote $p(X = x)$ as the probability that random variable X has value x (common abbreviation $p(x)$
- Discrete probabilities sum to 1 $\Sigma_x p(X = x) = 1$ and all probabilities are non- negative $p(X = x) \leq 0$

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Basics Concepts in Probability

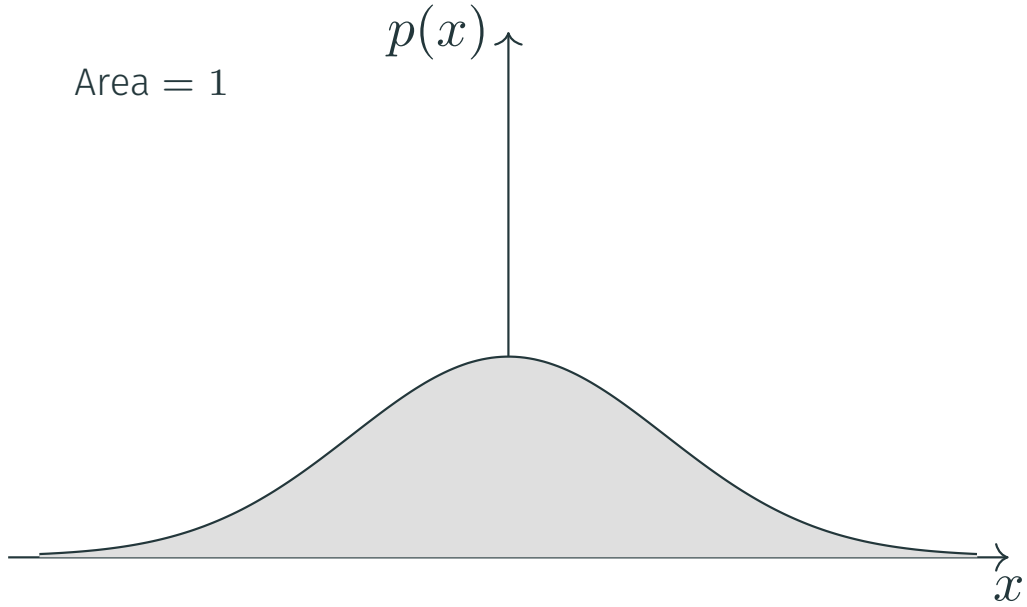
In robotics, many decision-making procedures happen in continuous space.

- Random variables take on a continuum of values according to probability density function (PDF)
- $p(x)$ measures *density per unit of* x, not probability, so it can take values that are greater than 1.
- The **area under the curve** must equal 1 (not the height of the curve).
- Rule $P(a \leq X \leq b) = \int_a^b p(x)\, dx$
- A PDF always integrates to 1

$$\int_{-\infty}^{\infty} p(x)\, dx = 1$$

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Basics Concepts in Probability



A common PDF is the normal distribution.

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Normal (Gaussian) Distribution

Definition:

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

- μ: mean (expected value)
- σ^2: variance (spread)
- Symmetric bell-shaped curve centered at μ
- $E[X] = \mu, \quad \mathrm{Var}(X) = \sigma^2$
- **Multi-variant:** Normal distributions over vectors:

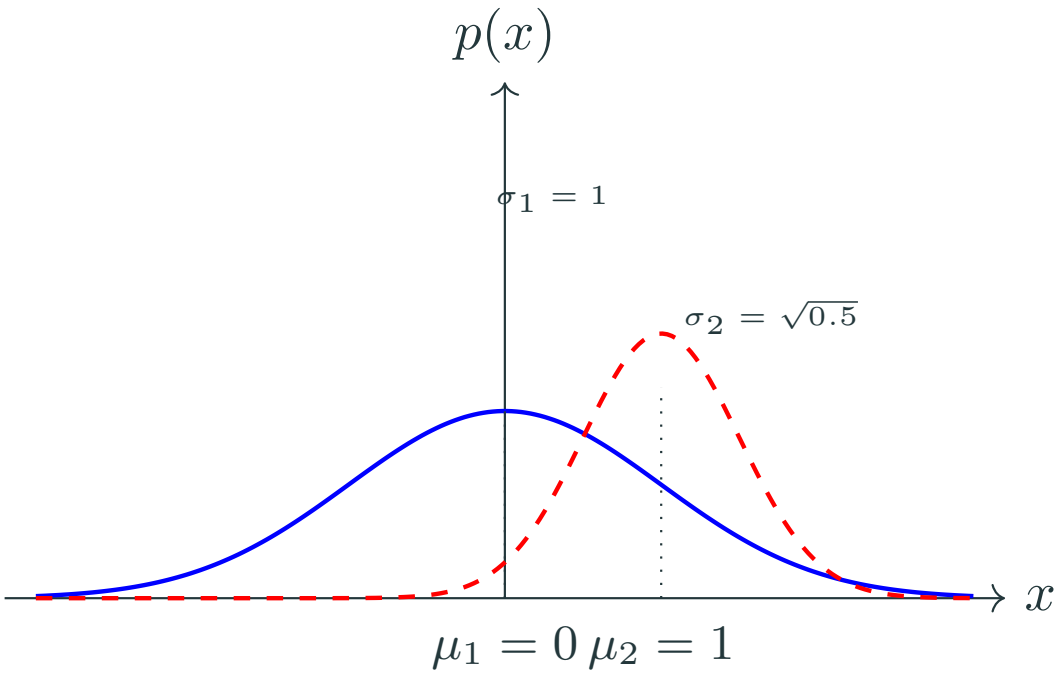
$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left[-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right]$$

Here, μ is the mean vector and Σ is the **covariance matrix**.

$$\int_{\mathbb{R}^n} p(\mathbf{x})\, d\mathbf{x} = 1$$

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Normal (Gaussian) Distribution



Solid: $\mu = 0, \sigma = 1$ Dashed: $\mu = 1, \sigma = 0.7$

In robotics, it is commonly used to model **sensor noise** and **motion uncertainty**.

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Joint and Conditional Distributions

Joint Distribution

$$p(x, y) = p(X = x, Y = y), \qquad \sum_x \sum_y p(x, y) = 1$$

Chain Rule:

$$p(x, y) = p(x \mid y)\, p(y)$$

Conditional Probability

$$p(x \mid y) = \frac{p(x, y)}{p(y)}, \qquad p(y) > 0$$

Independence

$$X \perp Y \iff p(x, y) = p(x)\, p(y)$$

Conditional Independence:

$$x \perp y \mid z \iff p(x, y \mid z) = p(x \mid z)\, p(y \mid z)$$

Even if X and Y are not independent marginally, they may be independent *given* z.

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Marginal Distributions and Independence

Marginal Distribution

- Discrete:

$$p(x) = \sum_y p(x, y), \qquad p(y) = \sum_x p(x, y)$$

- Continuous:

$$p(x) = \int p(x, y)\, dy, \qquad p(y) = \int p(x, y)\, dx$$

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Theorem of Total Probability

Discrete form:

$$p(x) = \sum_{y} p(x \mid y)\, p(y)$$

Continuous form:

$$p(x) = \int p(x \mid y)\, p(y)\, dy$$

This expresses the marginal distribution of X as a weighted average of its conditional distributions given Y.

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Bayes Rule

Discrete form:

$$p(x \mid y) = \frac{p(y \mid x)\, p(x)}{p(y)}, \qquad p(y) = \sum_i p(y \mid x_i)\, p(x_i)$$

Continuous form:

$$p(x \mid y) = \frac{p(y \mid x)\, p(x)}{p(y)}, \qquad p(y) = \int p(y \mid x')\, p(x')\, dx'$$

Since the denominator $p(y)$ does not depend on x it is often written as the normalizer

$$p(x \mid y) = \eta \quad p(y \mid x)\, p(x)$$

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Bayes Rule

- Plays a key role in robotics (and inference in general)
- If x is a quantity we would like to infer from y, the probability $p(x)$ is the **prior probability** and y is called the **data** (e.g., sensor measurement).
- The probability $p(x \mid y)$ is the **posterior probability**.
- The probability $p(y \mid x)$ is often called the **generative model**, since it describes how state variables X cause sensor measurements Y

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Expectation of a Random Variable

Definition:

- The **expectation** (or expected value) of a random variable X is the average value it takes under its probability distribution.

$$E[X] = \begin{cases} \sum_x x\, p(x), & \text{discrete} \\ \int x\, p(x)\, dx, & \text{continuous.} \end{cases},$$

Key Properties:

- Linearity: $E[aX + bY] = a\, E[X] + b\, E[Y]$
- If X and Y are independent, $E[XY] = E[X]\, E[Y]$

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Entropy

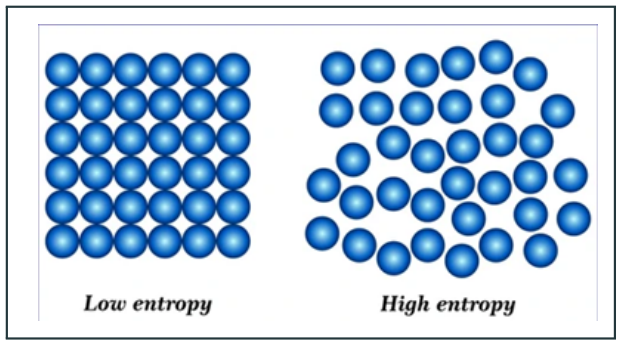
Measures the **uncertainty** or **information content** of a random variable. The entropy of a probability distribution

$$H_p(x) = E[-\log_2 p(x)]$$

which resolves to

$$H_p(x) = -\sum_x p(x)\ \log_2 p(x) \quad \text{(discrete case)}$$

$$H_p(x) = -\int p(x)\ \log p(x)\, dx \quad \text{(continuous case)}$$

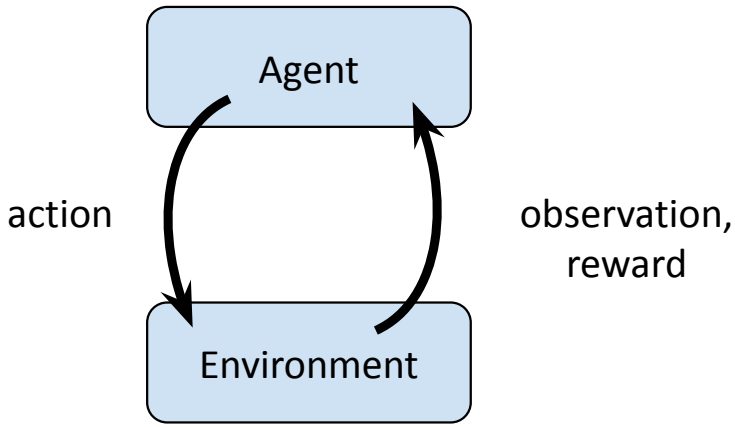


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Random Variables vs. Decision Variables

- When we explored probability theory, we discussed random variables that were set by probability distributions.
- When we consider an agent that needs to make decisions, we need to consider its decision variables



Key Idea: A **random variable** captures uncertainty. A **decision variable** captures choice.

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Random Variables vs. Decision Variables

Random Variables

- Represent quantities determined by **chance**.
- Not controlled by the agent.
- Described by a **probability distribution**
- Examples:
  - Sensor reading
  - Robot's true position
  - Next state
  - Reward

Decision Variables

- Represent quantities chosen by the **agent**.
- Controlled via a policy or a (deterministic or stochastic) control rule.
- Examples:
  - Control action
  - Grasp command `pick_up_apple`
  - Trajectory waypoint

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State, Action, Observation

- In AI, the state, action, and observation are marked as

$$s_t \ , \ a_t \ , \ o_t$$

- In robotics they are marked as

$$x_t \ , \ u_t \ , \ z_t$$

We will use these interchangeably

- In robotics, the dynamics are described by:
  - Motion model: $p(x_t \mid u_t, x_{t-1})$
  - Sensor model: $p(z_t \mid x_t)$

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From Probability to Markov Models

- Robots operate in uncertain environments evolving over time.
- A state is *complete* if it encodes all information necessary to predict the future (given action/control).
- **Markov assumption:**

$$p(x_t \mid x_{0:t-1}) = p(x_t \mid x_{t-1})$$

- The future depends only on the present state, not the full history.

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Definition: Markov Chain

Markov Chain
A sequence of random variables $X_0, X_1, \ldots$ is a Markov chain if

$$p(X_t = x_t \mid X_{0:t-1} = x_{0:t-1}) = p(X_t = x_t \mid X_{t-1} = x_{t-1})$$

- Transition model $p(x_t \mid x_{t-1})$
- Used for modeling stochastic dynamics.

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From MDPs to Markov Chains

MDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$

- The state X_t or S_t is a random variable, and the action U_t or A_t is a decision variable.
- The next state is set by the transition function / dynamics model $X_{t+1} \sim p(X_{t+1} \mid X_t, U_t)$

Markov Chain $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma \rangle$

- Agency not modeled
- Can be used when the policy is fixed or when agency is not explicitly modeled

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Transition Model and Evolution

- Transition probabilities often expressed as a matrix:

$$T_{ij} = p(x_t = j \mid x_{t-1} = i)$$

- Each row sums to 1.
- The belief (state distribution) evolves as:

$$p(x_t) = T^\top p(x_{t-1})$$

- Over time, Markov chains may converge to a stationary distribution π:

$$\pi = T^\top \pi$$

- π describes the long-term probability of each state.
- Useful for understanding steady-state behaviors in robotics.

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From Markov Chains to HMMs

- In many robotic systems, the true state x_t is not directly observable.
- We only perceive noisy measurements z_t.
- This leads to the **Hidden Markov Model (HMM)** which captures two random variables, the state and the observation:

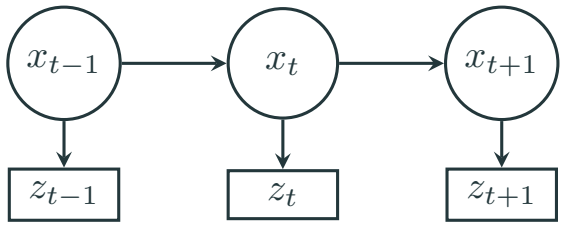
$$p(x_t \mid x_{t-1}), \quad p(z_t \mid x_t)$$

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Definition: Hidden Markov Model

$$\langle X, Z, p(x_0), p(x_t \mid x_{t-1}), p(z_t \mid x_t) \rangle$$

- X_t: hidden state
- Z_t: observation
- $p(x_0)$: prior distribution
- $p(x_t \mid x_{t-1})$: transition model
- $p(z_t \mid x_t)$: observation model



$$p(x_{0:T}, z_{1:T}) = p(x_0) p_{t=1}^{T} p(x_t \mid x_{t-1})\, p(z_t \mid x_t)$$

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From POMDPs to HMMs

POMDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, \Omega, \mathcal{O}, \beta_0 \rangle$

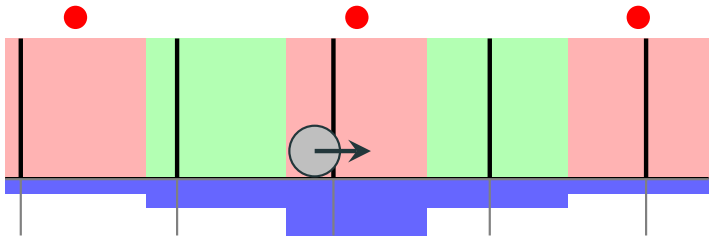
- Two random variables: state X_t and observation Z_t
- One decision variable: action U_t
- Dynamics model $X_{t+1} \sim p(X_{t+1} \mid X_t, U_t)$
- Sensor model $Z_{t+1} \sim p(Z_{t+1} \mid X_t, U_t)$

HMM $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma, \Omega, \mathcal{O}, \beta_0 \rangle$

- Can be used when the policy is fixed or when agency is not explicitly modeled.

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Example: 1D Robot Localization as an HMM



Belief distribution $\beta(x_t)$

- Hidden state x_t: robot position along hallway.
- Observation z_t: color of observed landmark.
- Belief combines motion and sensing uncertainty.

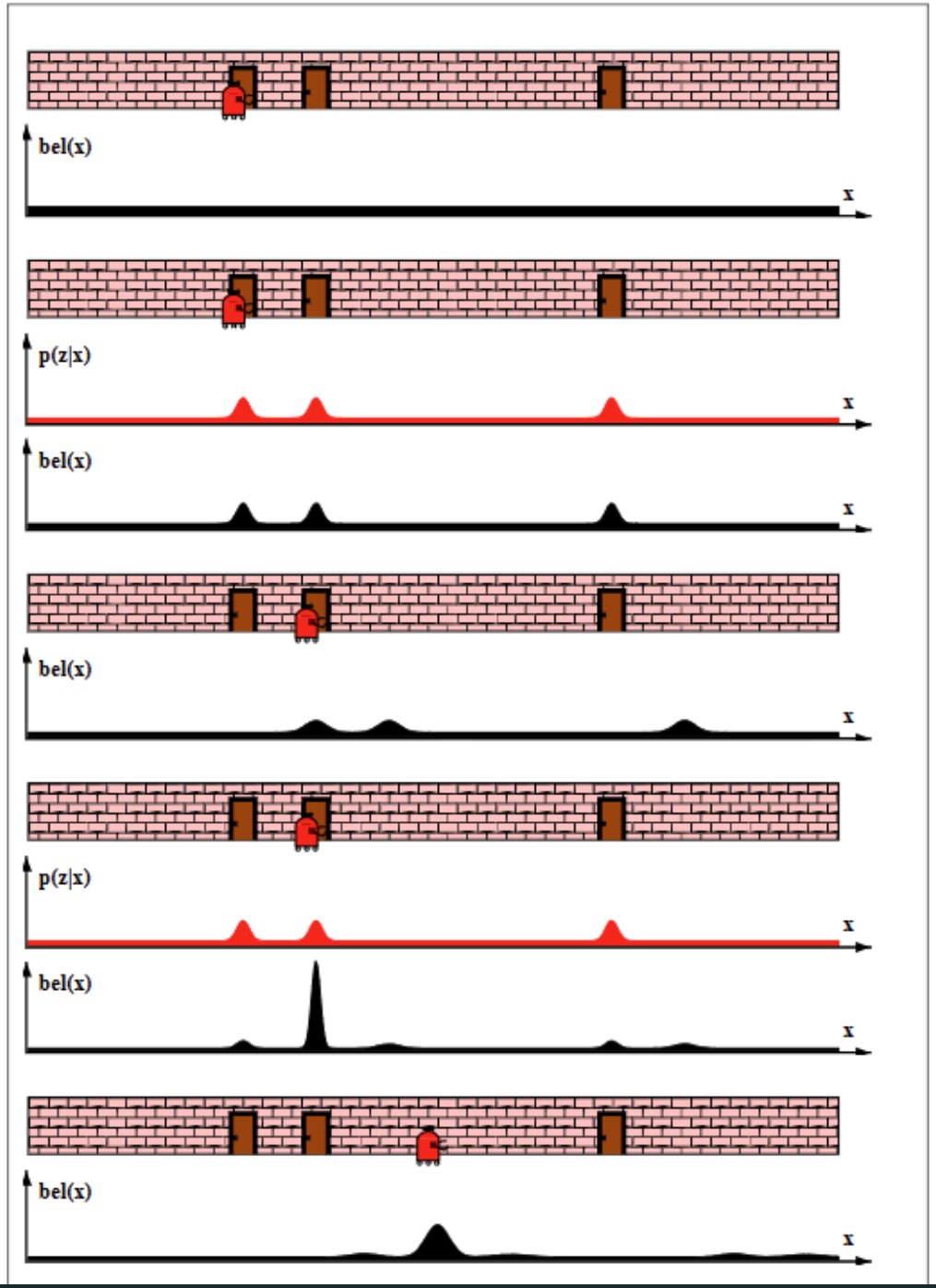
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Key Concepts

- State x_t, control/action u_t, sensor measurement z_t.
- Belief: $\beta(x_t) = p(x_t \mid z_{1:t}, u_{1:t})$.
- **Markov assumption:** given x_t, past and future are conditionally independent.
- Two-step update: prediction (motion) + correction (observation).

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bel(x)
x
p(z|x)
x
bel(x)
x
bel(x)
x
p(z|x)
x
bel(x)
x
bel(x)
x

Why recursive state estimation? Why maintain a belief?

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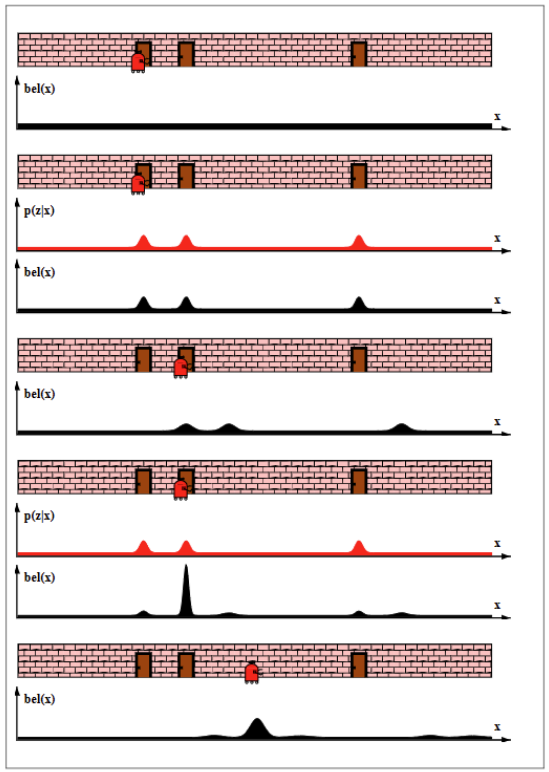
Why recursive state estimation? Why maintain a belief?

- Robots operate in uncertain, dynamic environments.
- Instead of relying on a “single best guess” state, we maintain distributions and make robust decisions.

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Intuition: Prediction vs. Correction

- **Prediction:** motion mode typically increases uncertainty.
- **Correction:** sensor model typically reduces uncertainty; shifts belief towards states consistent with measurement.
- **Underlying idea:** alternate these steps to yield an updated belief.



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Generative Model and Conditional Independence

Generative model for recursive state estimation

- x_t: hidden state (e.g., robot pose)
- u_t: control input (e.g., odometry)
- z_t: observation (e.g., sensor measurement)

Interpretation:

- The model describes how the world "generates" both the robot's motion and its observations over time.
- Each new state x_t depends only on the previous state x_{t-1} and control u_t.
- Each observation z_t depends only on the current state x_t.

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Generative Model and Conditional Independence

Generative Model for Recursive State Estimation

Conditional Independence Assumptions:

$$x_t \perp\!\!\!\perp \{x_{0:t-2}, z_{1:t-1}, u_{1:t-1}\} \mid (x_{t-1}, u_t),$$

$$z_t \perp\!\!\!\perp \{x_{0:t-1}, z_{1:t-1}, u_{1:t}\} \mid x_t$$

Consequence: this independence allows recursive estimation

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Recursive Bayesian Estimation

Starting point:

$$\beta(x_{t-1}) = p(x_{t-1} \mid z_{1:t-1}, u_{1:t-1})$$

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Recursive Bayesian Estimation

Starting point:

$$\beta(x_{t-1}) = p(x_{t-1} \mid z_{1:t-1},\, u_{1:t-1})$$

Prediction:

$$\underbrace{p(x_t \mid z_{1:t-1},\, u_{1:t})}_{\text{prior}} = \int p(x_t \mid u_t, x_{t-1})\ \underbrace{p(x_{t-1} \mid z_{1:t-1},\, u_{1:t-1})}_{\beta(x_{t-1})}\ dx_{t-1}$$

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Recursive Bayesian Estimation

Starting point:

$$\beta(x_{t-1}) = p(x_{t-1} \mid z_{1:t-1},\, u_{1:t-1})$$

Prediction:

$$\underbrace{p(x_t \mid z_{1:t-1},\, u_{1:t})}_{\text{prior}} = \int p(x_t \mid u_t, x_{t-1})\ \underbrace{p(x_{t-1} \mid z_{1:t-1},\, u_{1:t-1})}_{\beta(x_{t-1})}\ dx_{t-1}$$

Correction using observation:

$$\underbrace{\beta(x_t)}_{\text{posterior}} = \eta\, p(z_t \mid x_t)\ \underbrace{p(x_t \mid z_{1:t-1},\, u_{1:t})}_{\text{prior}}$$

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Bayes Filter

Algorithm `Bayes_filter`$(\beta(x_{t-1}), u_t, z_t)$

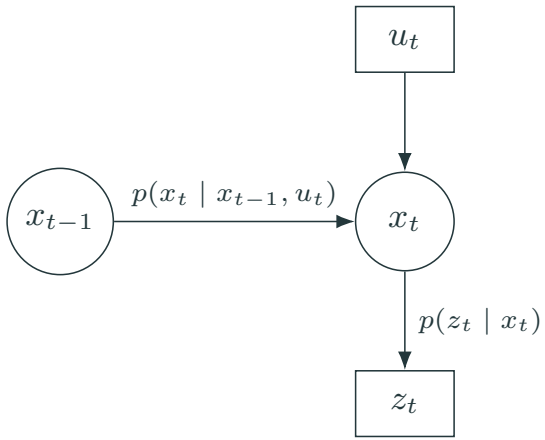
forall x_t do

$$\overline{\beta}(x_t) = \int p(x_t \mid u_t, x_{t-1})\, \beta(x_{t-1})\, dx$$

$$\beta(x_t) = \eta\, p(z_t \mid x_t)\, \overline{\beta}(x_t)$$

end

return $\beta(x_t)$



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Bayes filter

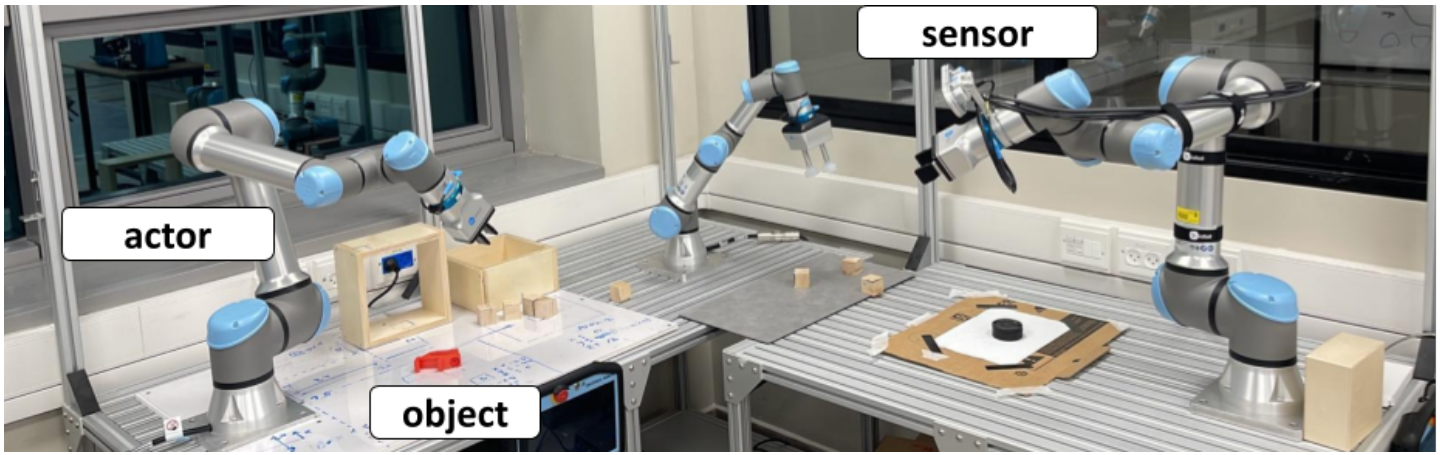
- The Bayes filter provides a generic recursion for the belief update.
- Specializations (e.g., Kalman filter, particle filter) will follow in later lectures.
- A critical foundation for localization, mapping, SLAM, etc.

How is it related to entropy?

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From Beliefs to Decisions



Value of Assistance for Grasping

Mohammad Masarwy, Yuval Goshen, David Dovrat, Sarah Keren

https://arxiv.org/abs/2310.14402

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Belief Update

When receiving an observation o, the agent updates its current belief β using its **belief update function** $\tau : \mathcal{B} \times \Omega \times \mathcal{S} \mapsto \mathcal{B}$ which maps belief $\beta \in \mathcal{B}$ to the new belief.

Commonly, a **Bayesian filter** is used:

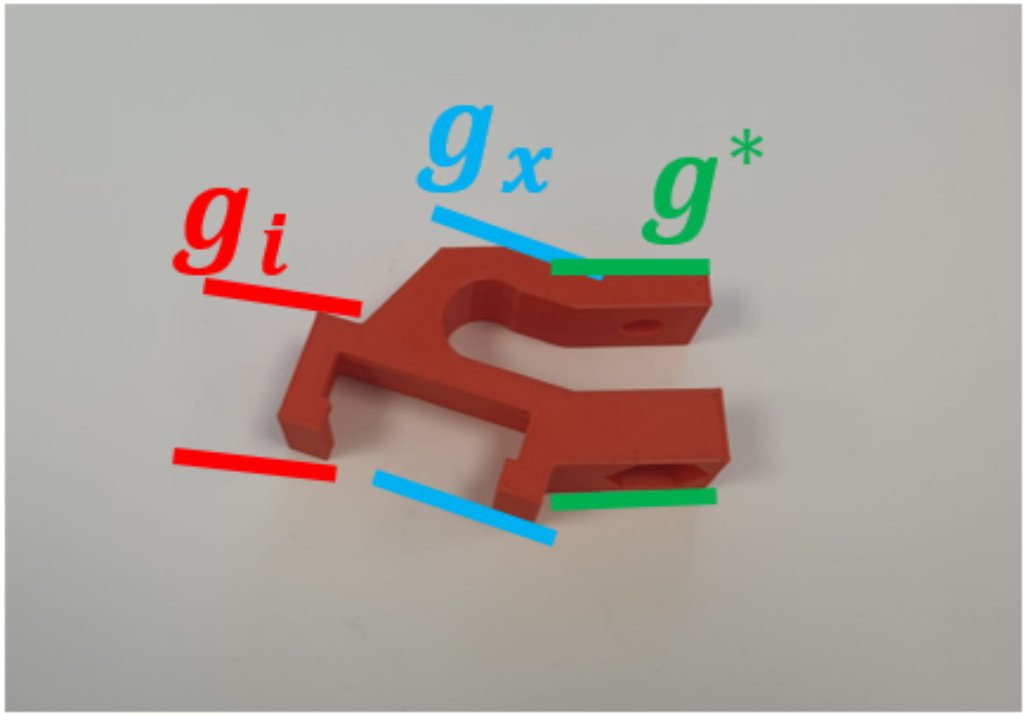
$$\beta^{o,a} = \frac{\mathcal{P}(o|s,a)\,\beta(s)}{\int_{s' \in \mathcal{S}} \mathcal{P}(o|s',a)\,\beta(s')\,ds'} \tag{1}$$

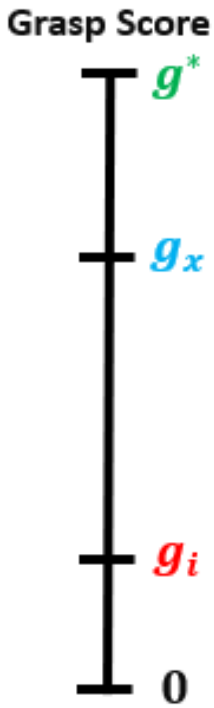
where $\mathcal{P}$ is the sensor function and $\beta(s)$ is the estimated probability that s is the actual world state
Discrete version:

$$\beta^{o,a} = \frac{\mathcal{P}(o|s,a)\,\beta(s)}{\Sigma_{p' \in \mathcal{P}_o} \mathcal{P}(o|s',a)\,\beta(s')} \tag{2}$$

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From Beliefs to Decisions





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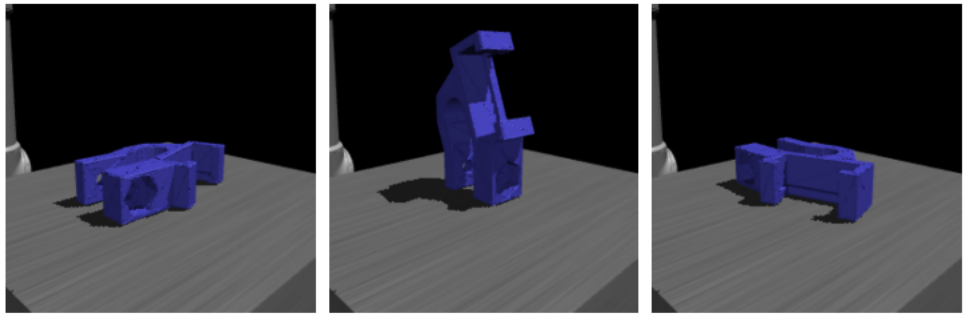
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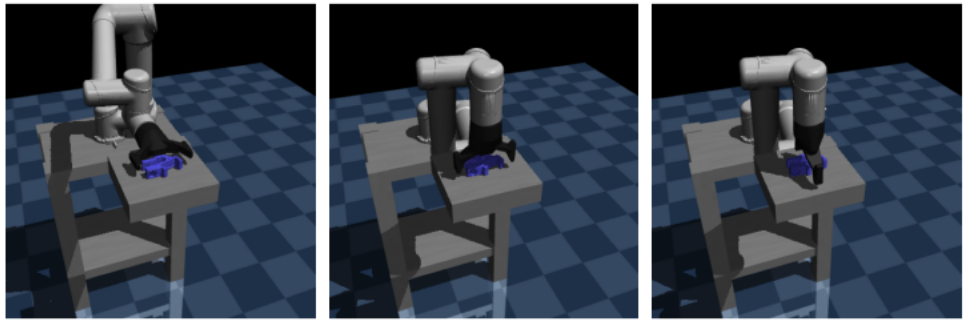
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From Beliefs to Decisions



(a) (b) (c)

Fig. 2: Example stable poses.



(a) (b) (c)

Fig. 3: Example grasp configurations from which the actor can attempt to grasp the object - each configuration is associated with a score, i.e., probability of success.

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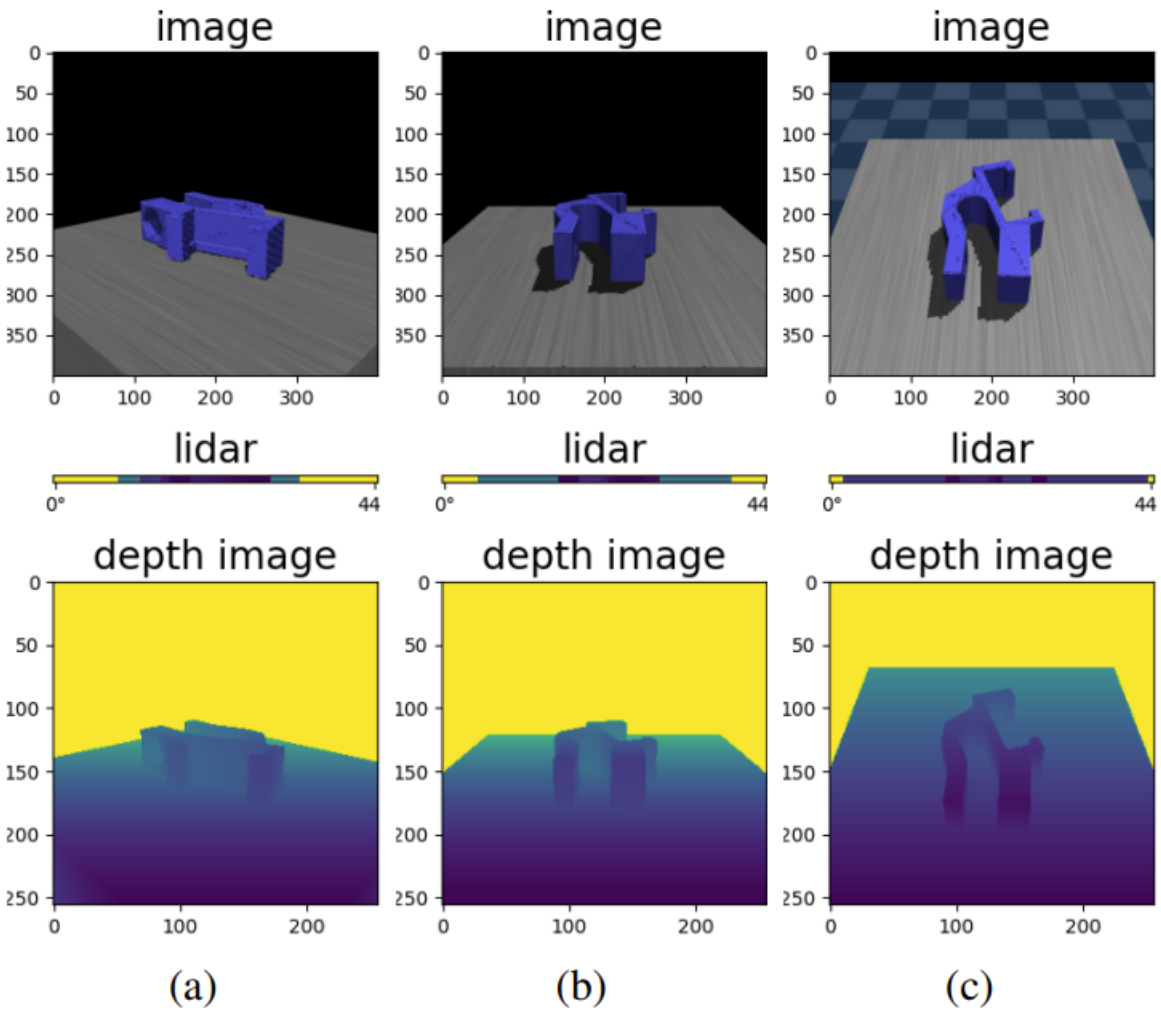


Fig. 4: Example sensor configurations and corresponding observations for a given stable pose. Each column represents the RGB image [top] lidar reading [middle] and depth image

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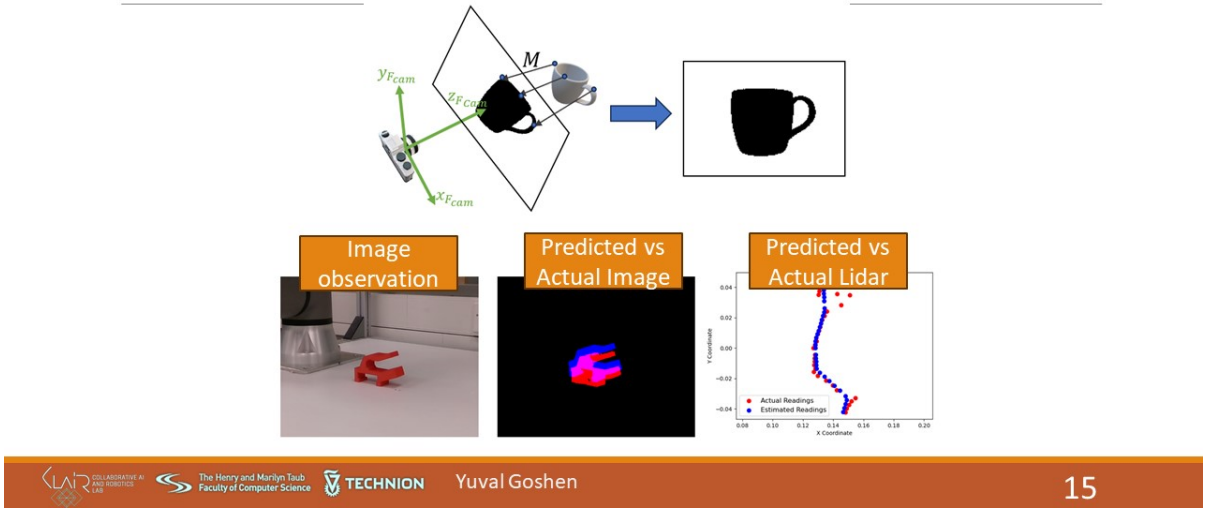
Definition (Sensor Function)

Given object pose $p \in \mathcal{P}_o$ and sensor configuration $q \in \mathcal{Q}$, sensor function $\mathcal{O} : \mathcal{P}_o \times \mathcal{Q} \mapsto \Omega$ is a random function, such that if $o = \mathcal{O}(p, q)$ then $P(o|p, q)$ provides the conditional probability of obtaining the observation o when the sensor configuration is q and the object pose is p.

Typically, the actual distribution is not known, and we use a *predicted sensor function* $\tilde{\alpha}$ and a *predicted observation probability* $\hat{P}$, which may be incorrect or inaccurate.

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Challenge: the probability of an observation is near 0.



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From Beliefs to Decisions

We use a similarity score, $\omega : \Omega \times \Omega \mapsto [0, 1]$ to compare the predicted and received observations:

$$\hat{P}(o|p, q) = \frac{\omega(\tilde{\alpha}(p, q), o)}{\int_{p' \in \mathcal{P}_o} \omega(\tilde{\alpha}(p', q), o) dp'} \tag{3}$$

The literature is rich of various definitions for ω, which may vary between applications and sensor types.

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From Beliefs to Decisions

For **belief update** we use a Bayesian filter such that for any observation $o \in \Omega$ taken from sensor configuration $q \in \mathcal{Q}$, the updated pose belief $\beta^{o,q}(p)$ for pose $p \in \mathcal{P}_o$ is given as

$$\beta^{o,q}(p) = \frac{\hat{P}(o|p,q)\,\beta(p)}{\int_{p' \in \mathcal{P}_o} \hat{P}(o|p',q)\,\beta(p')\,dp'} \tag{4}$$

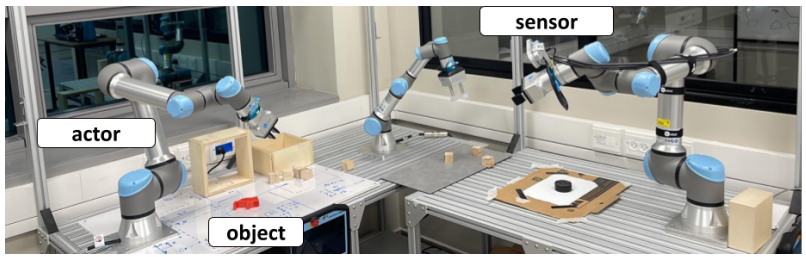
where $\beta(p)$ is the estimated probability that p is the object pose prior to considering the new observation o.

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Value of Assistance (VOA) for Grasping

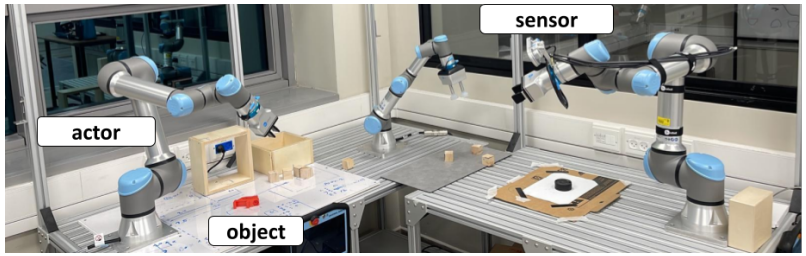
Given the actor's belief $\beta_a \in \mathcal{B}$, the belief of the helping agent $\beta_h \in \mathcal{B}$, the predicated observation probability $\hat{P}$, sensor configuration $q \in \mathcal{Q}$, and the actor's belief update function τ_a,

$$U_\alpha^{VOA}(\beta_h, \beta_{ac}) \stackrel{\text{def}}{=} \mathbb{E}_{p\sim\beta_h}\left[\mathbb{E}_{o\sim\hat{P}(o|p,q)}\left[\gamma(q_g^*(\beta_a^{o,q}), p)\right] - \gamma(q_g^*(\beta_{ac}), p))\right]. \tag{5}$$



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From Beliefs to Decisions





(a) P1 (b) P2 (c) P3 (d) P4 (e) P5 (f) P6

Is this a good observation?

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Summary

- 3-layered decision making in robotics
- Accounting for uncertainty in robotics
- Bayes filter

What next ?

- High-level task planning and low-level control
- Thinking about the right way to bridge them



What about sequential decision-making?

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